Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2343115	ADSA Lab	--	2	--	--	--	1	1

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II					
2343115	ADSA Lab	--	--	--	--	--	25	25	50

Lab Objective: The course aims to

Sr. No	Lab Objectives
1	To learn mathematical background for analysis of algorithm
2	To learn various advanced data structures.
3	To learn greed approach to solve problems.
4	To learn backtracking algorithm and maximum flow networks.
5	To learn dynamic programming methods.
6	To understand the concept of pattern matching.

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Lab Outcomes: Upon completion of the course students will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Understand methods for analysis of algorithms and solve recurrence problems.	L1, L2
2	Choose appropriate advanced data structures for a given problem and calculate its complexity.	L2, L3, L4
3	Analyze the greedy programming technique to solve problems.	L2, L3, L4, L5
4	Evaluate and analyze the backtracking algorithm and understand maximum flow networks.	L2, L3, L4, L5
5	Analyze the dynamic programming technique to solve problems.	L2, L3, L4, L5
6	Select a proper pattern matching algorithm for a given problem.	L3, L4, L5

Hardware & Software Requirements:

Hardware Requirement:	Software requirement:
PC i3 processor and above	Turbo/Borland C complier

DETAILED SYLLABUS:

Sr. No.	Name of Module	Suggested list of Practical	Hours	LO Mappings
I	**Introduction to Analysis of Algorithms**	1. Implement Merge sort and Quicksort for the given list of integer values and find space and time complexity. 2. Implementation of randomized quicksort algorithm and find space and time complexity. 3. Implementation of hash functions and its associated algorithms.	**04**	**LO1**
II	**Advanced Data Structures**	4. Construct Binary Search Tree for given sequence of integers and perform Pre-order, In-order and Post-order traversal of constructed tree. Analyze complexities. 5. Implement Insert, Delete, Search and Display operations on Binary Search Tree and analyze space and time complexity. 6. Implementation of operations on B/B+-trees.	**05**	**LO2**
III	**Greedy algorithms and Applications**	7. Implement solution for a 0-1 knapsack problem using dynamic programming. 8. Implement Prim's and Kruskal's algorithms. 9. Implement solution for job sequencing with deadlines problems.	**05**	**LO3**
IV	**Backtracking and Maximum flow Networks**	10. Implement N-Queen's problem using Back Tracking. 11. Implement Sum of subsets problem for a given set of distinct numbers using backtracking. 12. Implement graph coloring, 13. Implement Hamiltonian cycles. 14. Implementation of Ford-Fulkerson algorithm.	**04**	**LO4**
V	**Dynamic Algorithms**	15. Implementation of Bellman-Ford algorithm. 16. Implement Floyd Warshall Algorithm. 17. Implement Travelling salesman problem.	**04**	**LO5**
VI	**String Matching Algorithms**	18. Implement Naïve string matching algorithm. 19. Implement Rabin-Karp algorithm. 20. Implement KMP algorithm. 21. Implement Longest common subsequence. 22. Implement any one application.	**04**	**LO6**

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Text Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	Introduction to Algorithms	Cormen, Leiserson, Rivest, Stein	PHI	3rd Edition	2011
2	Algorithm Design	Jon Kleinberg, Éva Tardos	Pearson	1st Edition	2006
3	Data Structures and Algorithm Analysis in C++	Mark Allen Weiss	Pearson	4th Edition	2013

4	Introduction to the Design and Analysis of Algorithms	Anany Levitin	Pearson	3rd Edition	2011
5	Algorithms	Robert Sedgewick, Kevin Wayne	Addison-Wesley	4th Edition	2011

Online Resources:

S. No.	Website Name	URL	Modules Covered
1	NPTEL	https://archive.nptel.ac.in/courses/106/106/106106200/	M1
2	NPTEL	https://archive.nptel.ac.in/courses/106/105/106105085/	M2
3	NPTEL	https://archive.nptel.ac.in/courses/106/104/106104120/	M3
4	Coursera	https://www.coursera.org/learn/algorithms-part1	M1-M3
5	MIT OpenCourseWare	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-006-introduction-to-algorithms-fall-2011/	M1-M6

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Assessment :

Term Work: Term work shall consist of at least 10-12 practical's based on above list. Also Term work Journal must include at least 2 assignments.

Term Work Marks: 25 Marks (Total marks) = 15 Marks (Experiment) + 5 Marks (Assignments) + 5 Marks (Attendance)

Practical & Oral Exam: An Oral & Practical exam will be held based on Practicals.